

# Series and Approximations

## 1 Taylor Series

- expanding a function at a point
- power series arithmetic

## 2 Approximations

- truncated power series
- Padé approximations

MCS 320 Lecture 24  
Introduction to Symbolic Computation  
Jan Verschelde, 15 July 2026

# Series and Approximations

## 1 Taylor Series

- expanding a function at a point
- power series arithmetic

## 2 Approximations

- truncated power series
- Padé approximations

# Taylor Series

The *Taylor series* of  $f(x)$  at  $x = a$  is

$$f(a) + f'(a)(x - a) + f''(a)\frac{1}{2}(x - a)^2 + f'''(a)\frac{1}{3!}(x - a)^3 + \dots$$

or written with the derivative operator  $D$ , up to order  $n$

$$\sum_{k=0}^{n-1} \frac{D^k f(a)}{k!} (x - a)^k + O((x - a)^n),$$

where

- $D^k f(a)$  is the value of the  $k$ -th derivative of  $f(x)$  at  $x = a$ ,
- $O((x - a)^n)$  is the *order* of the series.

Application: the tangent line to a curve at a point is the Taylor series truncated after the linear term.

# Series and Approximations

## 1 Taylor Series

- expanding a function at a point
- power series arithmetic

## 2 Approximations

- truncated power series
- Padé approximations

# Power Series Arithmetic

We can multiply polynomials, but we cannot take their inverse.

Series with a nonzero constant term have a multiplicative inverse.

Example:

Given is  $a = a_0 + a_1x + a_2x^2 + \dots$ ,  
compute  $b = b_0 + b_1x + b_2x^2 + \dots$ , so  $a \cdot b = 1 + O(x^3)$ .

This is equivalent to solving a linear system in  $b_0, b_1, b_2$ :

$$a_0b_0 = 1,$$

$$a_0b_1 + a_1b_0 = 0,$$

$$a_0b_2 + a_1b_1 + a_2b_0 = 0.$$

If  $a_0 \neq 0$ , then  $b_0 = 1/a_0$  and the system has a unique solution.

# Series and Approximations

## 1 Taylor Series

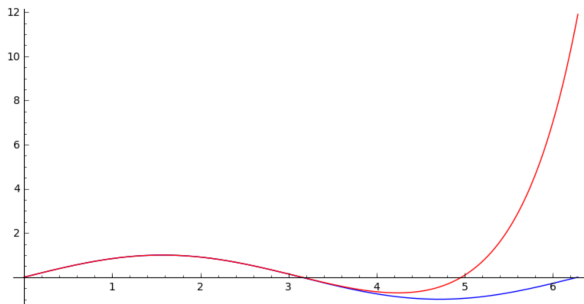
- expanding a function at a point
- power series arithmetic

## 2 Approximations

- truncated power series
- Padé approximations

# Truncated Power Series

Using a truncated power series to approximate  $\sin(x)$ :



Power series are good locally, but diverge globally.

# Series and Approximations

## 1 Taylor Series

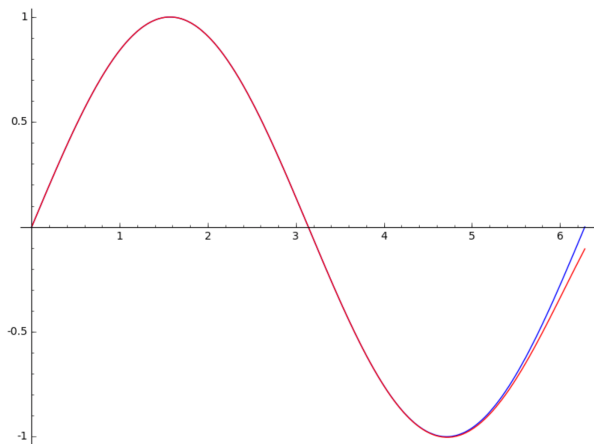
- expanding a function at a point
- power series arithmetic

## 2 Approximations

- truncated power series
- Padé approximations

# Padé Approximations

Using a rational approximation to approximate  $\sin(x)$ :



Rational approximations can achieve a global, uniform convergence.